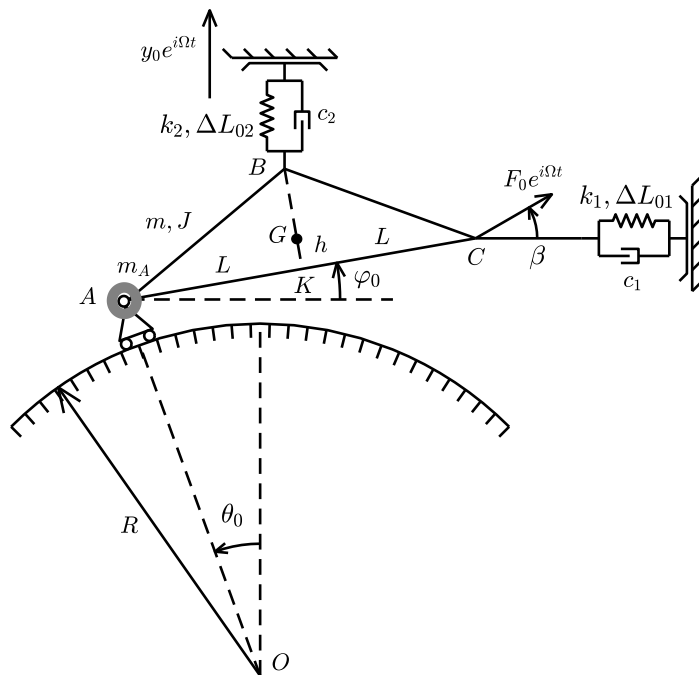


Dynamics of Mechanical Systems

July 01st, 2021



$m = 10 \text{ kg}$	$c_1 = 3 \text{ Ns/m}$
$m_A = 5 \text{ kg}$	$c_2 = 9 \text{ Ns/m}$
$J = 0.2 \text{ kgm}^2$	$\theta_0 = \pi/6$
$R = 1.0 \text{ m}$	$\varphi_0 = \pi/12$
$L = 0.25 \text{ m}$	$\beta = \pi/6$
$H = \overline{BK} = 0.15 \text{ m}$	
$h = \overline{GK} = 0.05 \text{ m}$	
$k_1 = 3000 \text{ N/m}$	
$k_2 = 9000 \text{ N/m}$	
$\Delta L_{01} = 0.0051 \text{ m}$	
$\Delta L_{02} = 0.0134 \text{ m}$	

The mechanical system above lies in the vertical plane and is shown in its equilibrium position. The ABC triangle is isosceles and the length of the base AC is $2L$. Point K is the midpoint of the base and the length of the height BK is H . The distance between the center of mass G and K is h . At point A there is a point mass m_A (depicted with a grey circle).

The students are requested to:

0. Write on top of each page their surname, name, ID number and SNxxx code (see below)
1. Write the linearized equations of motion around the given equilibrium configuration
2. Compute the natural frequencies and modal shapes;
3. Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$ inclined by the constant angle β . Output: the horizontal and the vertical component of the displacement of the center of mass G. Save the graphical results as SNxxx1.fig and SNxxx2.fig, with S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your person ID number (e.g. for Bruni Stefano 12345678: BS6781.fig and BS6782.fig)
4. Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic constraint displacement $y_0 e^{i\Omega t}$. Output: the elastic and damping force of the spring and damper system (k_2, c_2) and the force applied to the upper extremity of the spring and damper system (k_1, c_1) to get the given constraint displacement $y_0 e^{i\Omega t}$. Save the graphical results as SNxxx3.fig and SNxxx4.fig
5. Compute the Frequency Response Function (in the range $0 - 10 \text{ Hz}$, step = 0.01 Hz) applying the harmonic force $F_0 e^{i\Omega t}$ inclined by the constant angle β . Output: the constraint force in the cart A. Save the graphical result as SNxxx5.fig. Assuming a unilateral constraint model for the cart at point A, compute the maximum amplitude F_0 of the force that keeps the contact between the cart and the circular guide at the frequency equal to 4.0 Hz . Provide your answer in written, explaining the procedure used to derive the answer.
6. Compute the static preloads ΔL_{01} and ΔL_{02} considering a different configuration of the system in which $\theta_0 = \pi/3$, $H = \overline{BK} = 0.45 \text{ m}$, $h = \overline{GK} = 0.15 \text{ m}$ and $m = 20 \text{ kg}$.

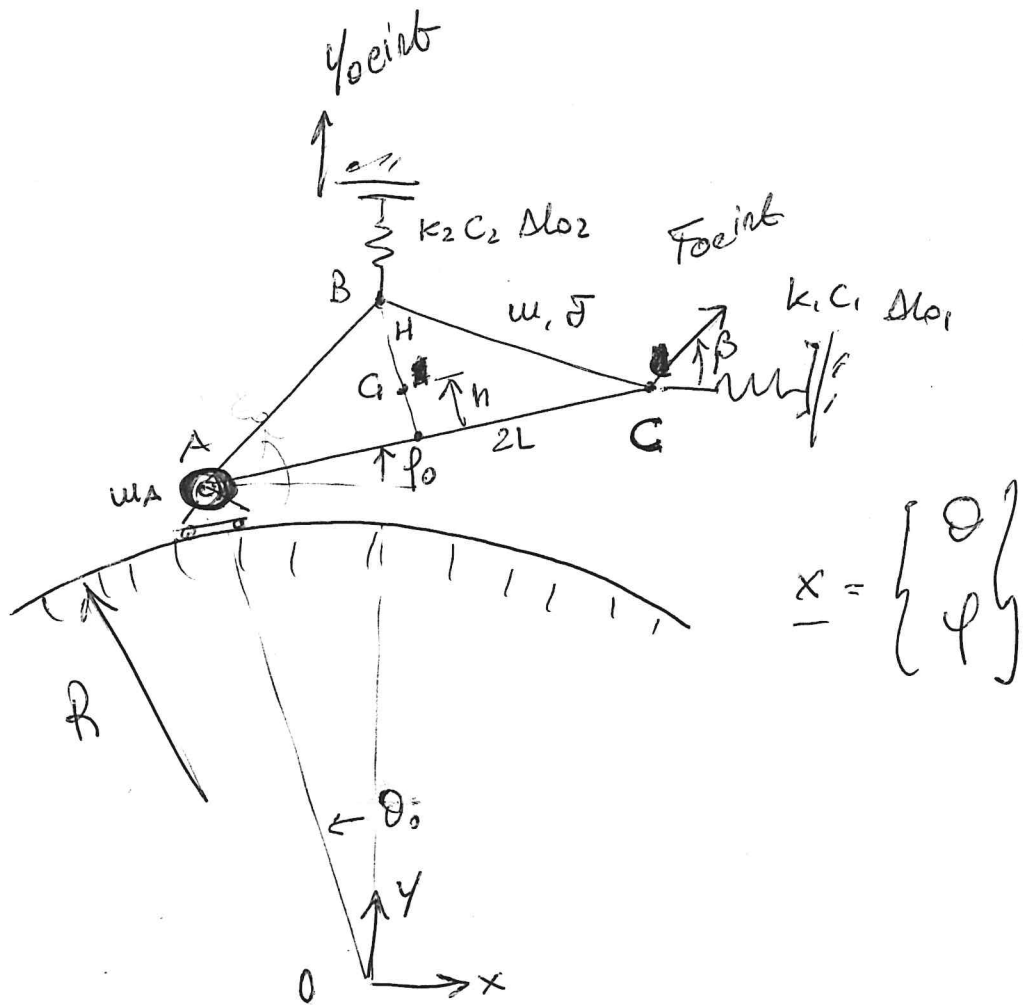
The answer to points 2 – 5 shall be provided by delivering the MATLAB script and .fig files as a compressed folder uploaded on BeeP, see instructions below.

Please be aware that the mark will be lowered by two points if:

- pdf files without the SNxxx code on top of the first page;
- the name of the zip file different from SNxxx.zip;
- the Beep field *Title* filled not accordingly to the indications given below at point 7.

Note: Please collect the Matlab's script file relative to the numerical solution of the exercise and save it with name SNxxx.m, with S the first letter of the surname, N the first letter of the name and xxx the last three digits of person ID number / codice persona (e.g. for Bruni Stefano 12345678: BS678.m). It is mandatory that all of the files (fig e .m) are collected in one single compressed folder (in .zip or.rar format) named SNxxx.zip (or SNxxx.rar), which has to be uploaded on Beep. To upload the sipped folder:

1. Connect to the site <https://beep.metid.polimi.it> and enter the portal using your credentials;
3. Select section **Homework** of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder **Test July 01st 2021 - standard**
5. Click on button Add and select Single document;
6. Browse the compressed file SNxxx.zip using the button Browse... (or Sfogliala...)
7. Fill the "Title" field with your code SNxxx (e.g. for Bruni Stefano 123456: BS456);
8. Confirm the procedure clicking the button Publish.



$$\underline{x} = \begin{Bmatrix} \theta \\ \phi \end{Bmatrix}$$

$$x_A = -R \sin \theta$$

$$y_A = R \cos \theta$$

$$x_G = -R \sin \theta + L \cos \phi - h \sin \phi$$

$$y_G = R \cos \theta + L \sin \phi + h \cos \phi$$

$$y_B = R \cos \theta + L \sin \phi + H \cos \phi$$

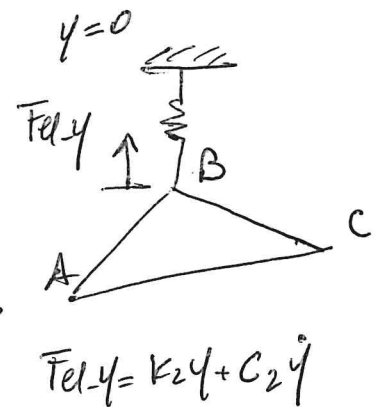
$$x_C = -R \sin \theta + 2L \cos \phi$$

$$y_C = R \cos \theta + 2L \sin \phi$$

$$T = \frac{1}{2} m_A \dot{V}_A^2 + \frac{1}{2} m \dot{x}_G^2 + \frac{1}{2} m \dot{y}_G^2 + \frac{1}{2} J \dot{\phi}^2$$

$$V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta l_i^2 ; D = \frac{1}{2} \sum_{i=1}^2 c_i \dot{\Delta l}_i^2$$

$$\delta W = \bar{F}_\theta \cos \beta \delta x_C + \bar{F}_\theta \sin \beta \delta y_C + \bar{F}_{el-y} \cdot \delta y_B$$



$$F_{el-y} = k_2 y + c_2 \dot{y}$$

$$\frac{\partial x_g}{\partial \theta} = -R \cos \theta \quad \frac{\partial x_g}{\partial \varphi} = -L \sec \varphi - h \cos \varphi$$

$$\frac{\partial y_g}{\partial \theta} = -R \sin \theta \quad \frac{\partial y_g}{\partial \varphi} = L \cos \varphi - h \sec \varphi$$

$$\frac{\partial y_b}{\partial \theta} = -R \sin \theta \quad \frac{\partial y_b}{\partial \varphi} = L \cos \varphi - H \sec \varphi$$

$$\frac{\partial x_c}{\partial \theta} = -R \cos \theta \quad \frac{\partial x_c}{\partial \varphi} = -2L \sec \varphi$$

$$\frac{\partial y_c}{\partial \theta} = -R \sin \theta \quad \frac{\partial y_c}{\partial \varphi} = 2L \cos \varphi$$

$$A_{uz} = \begin{bmatrix} -R \cos \theta_0 & -L \sec \varphi_0 - h \cos \varphi_0 \\ -R \sin \theta_0 & L \cos \varphi_0 - h \sec \varphi_0 \\ 0 & 1 \end{bmatrix} \begin{matrix} x_c \\ y_c \\ \varphi \\ s_A \end{matrix}$$

$$A_{kc} = \begin{bmatrix} R \sin \theta_0 & -L \cos \varphi_0 + H \sec \varphi_0 \\ R \cos \theta_0 & 2L \sec \varphi_0 \end{bmatrix}$$

$$A_e = \begin{bmatrix} -R \cos \theta_0 & -2L \sec \varphi_0 \\ -R \sin \theta_0 & 2L \cos \varphi_0 \\ -R \sin \theta_0 & L \cos \varphi_0 - H \sec \varphi_0 \end{bmatrix} \begin{matrix} F_0 \cos \varphi \\ F_0 \sec \varphi \\ k_2 y + c_2 \dot{y} \end{matrix}$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = R \cos \theta \quad \frac{\partial^2 \Delta L_2}{\partial \theta \partial \phi} = 0 \quad \frac{\partial^2 \Delta L_2}{\partial \phi^2} = L \sec \phi + H \cos \phi$$

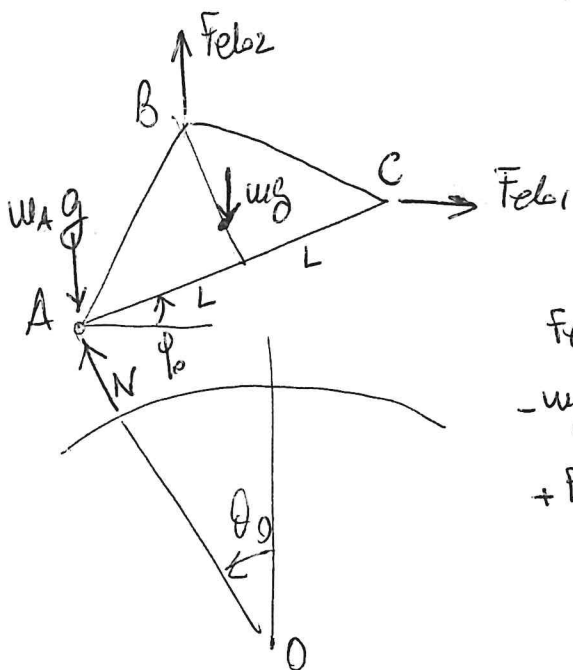
$$\frac{\partial^2 \Delta L_1}{\partial \theta^2} = -R \sec \theta \quad \frac{\partial^2 \Delta L_1}{\partial \theta \partial \phi} = 0 \quad \frac{\partial^2 \Delta L_1}{\partial \phi^2} = 2L \cos \phi$$

$$[K_{EL,1}] = k_1 \Delta L_1 \begin{bmatrix} -R \sec \theta_0 & \phi \\ \phi & 2L \cos \phi_0 \end{bmatrix}$$

$$[K_{EL,2}] = k_2 \Delta L_2 \begin{bmatrix} R \cos \theta_0 & 0 \\ 0 & L \sec \phi_0 + H \cos \phi_0 \end{bmatrix}$$

$$\frac{\partial^2 \psi_4}{\partial \theta^2} = -R \cos \theta \quad \frac{\partial^2 \psi_4}{\partial \theta \partial \phi} = 0 \quad \frac{\partial^2 \psi_4}{\partial \phi^2} = -L \sec \phi - h \cos \phi$$

$$k_5 = w_g \begin{bmatrix} -R \cos \theta_0 & 0 \\ 0 & -L \sec \phi_0 - h \cos \phi_0 \end{bmatrix} \quad [K_{SA}] = w_A g \begin{bmatrix} -h \cos \theta_0 & \phi \\ \phi & \phi \end{bmatrix}$$



$$\sum M_A = 0$$

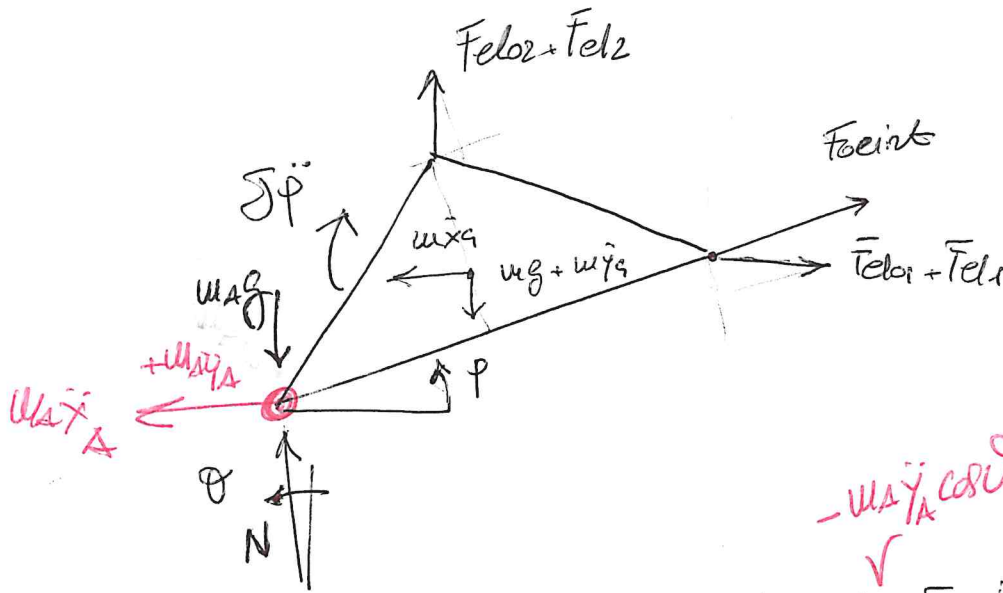
$$F_{EL2} (L \cos \phi_0 - H \sec \phi_0) - w_g (L \cos \phi_0 - h \sec \phi_0) + F_{EL1} 2L \sec \phi_0 = 0$$

$$\sum M_B = 0$$

$$F_{EL2} (R \sec \theta_0 - L \cos \phi_0 + H \sec \phi_0) + w_g (R \sec \theta_0 - L \cos \phi_0 + h \sec \phi_0) + F_{EL1} (R \cos \theta_0 + 2L \sec \phi_0) = 0$$

$$-w_A g R \sec \theta_0$$

$$= \begin{bmatrix} -2L \sin \phi_0 & L \cos \phi_0 - h \sin \phi_0 \\ R \cos \phi_0 + 2L \sin \phi_0 & R \sin \phi_0 - L \cos \phi_0 + h \sin \phi_0 \end{bmatrix} \begin{Bmatrix} F_{el1} \\ F_{el2} \end{Bmatrix} =$$



$$N + F_{el2} \cos \vartheta + F_{el2} \cos \vartheta - m g \cos \vartheta - m \ddot{x}_s \cos \vartheta + F_{el}^{int} \sin \vartheta \cos \vartheta + \underbrace{- m_A g \cos \vartheta}_{=0} - F_{el1} \sin \vartheta - F_{el1} \sin \vartheta - F_{el}^{int} \cos \vartheta \sin \vartheta + m \ddot{x}_s \sin \vartheta = 0$$

$$N_{ST} = \text{uf Costo} - \text{Felo2 Costo} + \text{Felo1 sendo} + \text{uag Costo}$$

$$N_d = (F_{el2} \sin \vartheta_0 - m g \sin \vartheta_0 + F_{el1} \cos \vartheta_0)(1 - \vartheta_0) - (F_{el2} \cos \vartheta_0 + F_{el1} \sin \vartheta_0) \cos \vartheta_0 + (F_{el1} + F_{el2} \cos \vartheta_0 - m \ddot{x}_a) \sin \vartheta_0$$

$$+ (F_{el_1} + F_{el_2} \cos \beta - m \ddot{x}_a) \sin \theta_0$$

- Mag seeds

$$V = K_1 (\Delta l_{01} + \Delta l_1)^2 + K_2 (\Delta l_{02} + \Delta l_2)^2 + m g h_S + m_A g h_A$$

$$\frac{\partial V}{\partial \theta} = K_1 (\Delta l_{01} + \Delta l_1) \frac{\partial \Delta l_1}{\partial \theta} + K_2 (\Delta l_{02} + \Delta l_2) \frac{\partial \Delta l_2}{\partial \theta} + m g \frac{\partial h_S}{\partial \theta} + m_A g \frac{\partial h_A}{\partial \theta}$$

$$\frac{\partial V}{\partial \phi} = \dots + m_A g \frac{\partial h_A}{\partial \phi}$$

$$\begin{cases} K_1 \Delta l_{01} \frac{\partial \Delta l_1}{\partial \theta} + K_2 \Delta l_{02} \frac{\partial \Delta l_2}{\partial \theta} + m g \frac{\partial h_S}{\partial \theta} \Big|_{x_0} = 0 \\ K_1 \Delta l_{01} \frac{\partial \Delta l_1}{\partial \phi} \Big|_{x_0} + K_2 \Delta l_{02} \frac{\partial \Delta l_2}{\partial \phi} \Big|_{x_0} + m g \frac{\partial h_S}{\partial \phi} \Big|_{x_0} + m_A g \frac{\partial h_A}{\partial \phi} \Big|_{x_0} = 0 \end{cases}$$

$$\begin{bmatrix} \frac{\partial \Delta l_1}{\partial \theta} \Big|_{x_0} & \frac{\partial \Delta l_2}{\partial \theta} \Big|_{x_0} \\ \frac{\partial \Delta l_1}{\partial \phi} \Big|_{x_0} & \frac{\partial \Delta l_2}{\partial \phi} \Big|_{x_0} \end{bmatrix} \begin{bmatrix} K_1 \Delta l_{01} \\ K_2 \Delta l_{02} \end{bmatrix} = \begin{bmatrix} - \frac{\partial h_S}{\partial \theta} \Big|_{x_0} \\ - \frac{\partial h_S}{\partial \phi} \Big|_{x_0} \end{bmatrix} m g - \begin{bmatrix} \frac{\partial h_A}{\partial \theta} \\ \frac{\partial h_A}{\partial \phi} \end{bmatrix} m_A g$$

$$\begin{bmatrix} R \cos \phi_0 & R \sin \phi_0 \\ 2L \sin \phi_0 & -L \cos \phi_0 + H \sin \phi_0 \end{bmatrix} \begin{bmatrix} K_1 \Delta l_{01} \\ K_2 \Delta l_{02} \end{bmatrix} = \begin{bmatrix} R \sin \phi_0 \\ -L \cos \phi_0 + H \sin \phi_0 \end{bmatrix} m g + \begin{bmatrix} R \sin \phi_0 \\ \phi \end{bmatrix} m_A g$$

Computation of the springs preloads
zeroing the first partial derivatives
of the potential energy

$$(N \cos \phi = N_1 \cos \phi_0 - N_2 \sin \phi_0 (\phi - \phi_0) = m \ddot{y}_S + m_A \ddot{y}_A - F_{el2} - F_0 e^{\text{int}} \sin \phi)$$

↑
not important for the solution